

Computational Interaction Science Concept-Source Matrix

Auditable evidential basis for canonical concepts

Concept	Primary sources	Supporting sources	Current status
Interaction	(1982) Human-computer interaction: a preliminary RUDD,J., STERN, K. R. AND ISENSEE, S. (1996) Low social psychological analysis. Behavior Research: vs. High-fidelity Prototyping Debate. ACM Inter- Methods and Instrumentation,13, (2), 210-220. actions Magazine, January, 76-85. References 501 (1993) Human-Computer Interaction (2nd ed.). Analysis: Verbal Reports as Data. Cambridge, MA: London: Prentice-Hall Europe. The MIT Press. (1993) Project Ernestine: validating a GOMS J. (1993) Unpacking collaboration: the interac- analysis for predicting and explaining real-world tional organization of trading in a city dealing performance. Human-Computer Interaction, 8(3), room. In Proceedings of the Third European Con- 237-309. ference on Computer-Supported Cooperative 1. Interview: reproduced by permission of IDEO. (1997) Tangible Bits: towards seamless interfaces, in CH1'97 Proceedings, reprinted by permission of Chapter 2 Association for Computing Machinery, Inc.; Figure Figure 2.1 (on Color Plate 2): gif from www.ai.mit. 2.1 1 (ii) gif from www.almaden.ibm.com/cs/blueeyes/edu/projects/medical-vision/surgery/ reproduced by permission of IBM; Figure 2.11 (iii): gif surgical-navigation.htm1	5.4(a)and (b): Marcus, A. (1993) Human and Hix, D. (1989) How Microsoft builds software, communication in advanced uls, Figures 2 and 4 International Journal of Man-Machine Studies, 31, (pages 106 and 107) in Communications of the ACM, 477-494, the Star lifecycle model, reproduced by 36(4), 101-109, reprinted by permission of permission of Academic Press Ltd.; Figure 6.14: The Association for Computing Machinery, Inc.; Figure usability engineering lifecycle figure in The Usability A simplest systematics for the organization of SHNEIDERMAN, B. (1998) Designing the User Inter- turn-taking for conversation. Language, 50, face: Strategies for Effective Human-Computer 696-735. Interaction (3rd ed.). Reading, MA: Addison- Abramsky, S. (2013). Semantics of interaction. arXiv:1312.0121. and blind people. In Proceedings of ICCHP '92, GAVAR, B., DUNNE, T., AND PACENTI, E. (1999) Cul- Vienna: Austrian Computer Society, 114-119. tural probes. ACM Interactions Magazine, January AND SCHOONARD, J. (1987) The 1984 Olympic cally derived methodologies and tools for human- Message System: a test of behavioral principles of computer interface	Established synthesis

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Representation and projection	5.4(a) and (b): Marcus, A. (1993) Human and Hix, D. (1989) How Microsoft builds software, communication in advanced users, Figures 2 and 4 International Journal of Man-Machine Studies, 31, (pages 106 and 107) in Communications of the ACM, 477-494, the Star lifecycle model, reproduced by 36(4), 101-109, reprinted by permission of Academic Press Ltd.; Figure 6.14: The Association for Computing Machinery, Inc.; Figure usability engineering lifecycle figure in The Usability AND MADACHY, R. (1998) Using the WinWin spiral Research. In Proceedings of the Virtual Worlds real model: a case study. IEEE Computer, 31(7), Conference 2000, Paris, France. AND SCHOONARD, J. (1990) The 1984 Olympic models by extended traceability to captured system Message System: a test of behavioral principles of usage. Interacting with Computers, 13(1), 77-95. ATKINSON, P., AND HAMMERSLEY, M. (1994) Ethnographic requirements. In J. Bennett, D. Case, J. Sandelin, Ruffy and participant observation. In N. K. Denzin and M. Smith (eds.) Visual Display Terminals: Usability and Y. S. Lincoln (eds.) Handbook of Usability Issues and Health Concern. Englewood Qualitative Research. London: Sage. Cliffs, NJ:	B. (1999) (eds.) Readings in Information Visualization CUSUMANO, M. A. AND SELBY, R.W. (1997) How Microsoft builds software. Communications of the ACM, 40(6), 53-61. Baier, C., & Katoen, J.-P. (2008). Principles of Model Checking. MIT Press. Baier, C., & Katoen, J.-P. (2008). Principles of Model Checking. MIT Press. Comparative Interaction 10 BELLOTTI, V AND ROGERS, Y. (1997) From web press Interactions Magazine, January and February, to web pressure: multimedia representations and 25-30. 494 References	Established synthesis

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Measurement	(1995) Bifocal tools for scenarios and representative Evaluating the usability of an evolving collaborations in participatory activities with users. In J. M. tive product--changes in user type, tasks and evaluation methods over time. In Proceedings of John Wiley & Sons, 135-163. GROUP'99,150-159. (1999) MEDLINEplus Interface Evaluation: Final electronic teller of stories, Robots for Kids (Druin, Report, reproduced by permission of Professor Keith C.A. and Helander, J., eds.) reproduced by Cogdill, College of Information Studies, University permission of Academic Press, Inc. and the authors, of Maryland; Figure 14.2: Figure 2, pages 67-80, Institute for Advanced Computer Studies, University from Lund, A.M. Ameritech's usability laboratory: of Maryland. from prototype to final design, Behaviour and (2000) PETS: A personal electronic teller of stories. NIELSEN, J. (1992) Finding usability problems through tries.' In C.A. Druin & J. Helander (eds.) Robots heuristic	1. Interview: reproduced by permission of IDEO. (1997) Tangible Bits: towards seamless interfaces, in CH1'97 Proceedings, reprinted by permission of Chapter 2 Association for Computing Machinery, Inc.; Figure Figure 2.1 (on Color Plate 2): gif from www.ai.mit. 2.1 1 (ii) gif from www.almaden.ibm.com/cs/blueeyes/edu/projects/medical-vision/surgery/ reproduced by permission of IBM; Figure 2.11 (iii): gif surgical-navigation.htm1 reproduced by permission from www.parc.xerox.com/red/members/richgold/ of Michael E. Leventon; Figure 2.6(a): gif from livingdoc/slide6.html, reproduced by permission of http://vibes.cs.uiuc.edu/ProjectNRIVirtueVirtueOve Rich Gold of PARC Communications; Figure 2.12: rview.htm, reproduced by permission of Dr Daniel A. gif from www.mbay.net/-brendaWarticles/IPDA. Reed (University of Illinois at Urbana-Champaign) Mar.951 reproduced by permission of General Magic, from work on the Collaborative Virtual Inc.;	Established synthesis

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Similarity and equivalence	Abate, A., Bacci, G., Bacci, G., & Larsen, K. G. (2017). On the relationship between bisimulation and trace equivalence in an approximate probabilistic context. García-Bañuelos, L., van Beest, N. R. T. P., Dumas, M., La Rosa, M., & Mertens, W. (2018). Complete and interpretable conformance checking of business processes. IEEE Transactions on Software Engineering, 44(3), 262-290. Gower, J. C. (1971). A general coefficient of similarity and some of its properties. Biometrics, 27, 857-871. Lange, M., Lozes, E., & Vargas Guzman, M. (2013). Model-checking process equivalences. Theoretical Computer Science, 560, 326-347.	Park, D. (1981). Concurrency and automata on infinite sequences. In Theoretical Computer Science. Springer. Page 12 Interaction Similarity Syring, A. F., Tax, N., & van der Aalst, W. M. P. (2019). Evaluating conformance measures in process mining using conformance propositions. Tversky, A. (1977). Features of similarity. Psychological Review, 84(4), 327-352. van der Aalst, W. M. P. (2016). Process Mining: Data Science in Action (2nd ed.). Springer. van der Aalst, W. M. P. (2016). Process Mining: Data Science in Action. Springer.	Established synthesis
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